

Yuyue Zhang  
China University of Petroleum (East China)  
Email: 15044682275@163.com  
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The Editor  
*Knowledge-Based Systems*

Dear Editor,

We are pleased to submit our original research article entitled “*Knowledge Lifecycle Management for Quantitative Trading: A Knowledge-Based Expert System with Contrastive State Discovery and Hierarchical Routing*” for consideration for publication in *Knowledge-Based Systems*.

## Background and Problem

Quantitative trading systems face a critical but underexplored challenge: when market regimes shift, predictive knowledge is abruptly lost. Existing approaches either train a single model across all regimes—which fails to adapt—or maintain independent experts—which cannot share cross-state knowledge. This “knowledge forgetting” problem has been largely overlooked in the financial machine learning literature.

## Our Contribution

To address this challenge, we propose a *Knowledge Lifecycle Framework* with three integrated mechanisms:

1. **Contrastive state discovery** that learns latent market state structures from data without manual heuristics, enabling adaptive state identification;
2. **A hierarchical architecture** with a shared encoder and lightweight state adapters that preserves cross-state knowledge while enabling rapid specialization, using 68.7% fewer parameters than independent experts;
3. **State ambiguity awareness** and **bidirectional knowledge flow** that dynamically adjust expert routing based on identification confidence and enable expert-specific knowledge to enrich the shared encoder.

## Key Findings

We validate our framework on 18 stocks over 5.5 years with strict rolling window backtesting and 5 repeated runs. Our system achieves:

- Positive Sharpe ratios on 16/18 stocks (88.9%), with permutation test  $p = 0.0006$  confirming statistical significance;
- Cross-stock Sharpe standard deviation of 0.55, substantially lower than all baseline and state-of-the-art models;
- 42.3% reduction in knowledge forgetting compared to independent experts;
- Successful zero-shot generalization from A-shares to US equities with only 6.2% accuracy degradation.

At the equal-weight portfolio level, our system achieves Sharpe 2.87 and outperforms all baseline strategies.

### **Why This Work Fits *Knowledge-Based Systems***

This work is designed from the ground up as a practical knowledge-based expert system. We explicitly formalize market states as *knowledge states*—each encapsulating patterns, rules, and heuristics about price behavior under specific conditions. Our system represents knowledge at two levels (general knowledge via shared encoder, state-specific knowledge via adapters), and the expert routing mechanism is interpreted as a *knowledge retrieval and inference process*. This aligns directly with the journal’s core themes: knowledge representation, knowledge retrieval, and knowledge-based reasoning for decision support. Furthermore, we provide a fully open-source interactive deployment terminal that visualizes the entire decision pipeline, bridging the gap between algorithmic research and practical applications.

### **Originality and Significance**

To the best of our knowledge, this is the first work to:

- Formalize a Knowledge Lifecycle Framework (acquisition → preservation → update → forgetting) for financial time series prediction;
- Demonstrate bidirectional knowledge flow between shared and expert-specific components in a trading system;
- Quantify knowledge forgetting rates across market states and show they satisfy a metric structure (triangle inequality).

### **Declarations**

We confirm that:

- This manuscript has not been published previously and is not under consideration elsewhere;
- All authors have read and approved the final manuscript;
- There are no competing financial or non-financial interests to declare;
- Generative AI tools (Grammarly and ChatGPT) were used only to improve language fluency and assist with literature organization; all content was reviewed and edited by the authors, who take full responsibility for the published article.

We believe this work will be of significant interest to the readership of *Knowledge-Based Systems* and would be grateful for your consideration.

Thank you for your time and consideration. We look forward to hearing from you.

Yours sincerely,

Yuyue Zhang  
Corresponding Author  
China University of Petroleum (East China)

Email: 15044682275@163.com